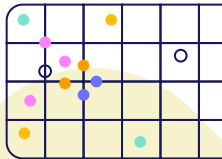
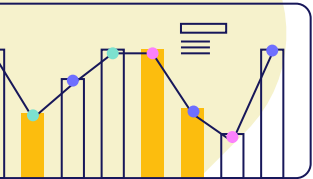


High School Students Breakfast Habits Study

Amanda Ruiz, Juliana Guerra, & Natalia de
Armas





Question

RQ: Is the proportion of students at Doral Academy High School who eat breakfast every school day greater than the national average of 27.4%?

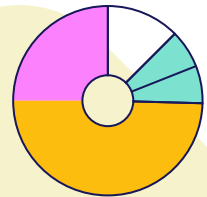
$$H_o: p = 0.274$$

$$H_a: p > 0.274$$

p = the population proportion of high school students in Doral Academy who eat breakfast daily before school.

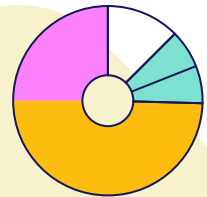
Literature Review

- **The Center for Disease Control and Prevention**
- Representative sample of U.S. high school students to examine health-related behaviors, including breakfast consumption
- Findings from the survey showed that only **27.4%** of U.S. high school students eat breakfast every day before school
- Examined connections between skipping breakfast and academic grades, feelings of sadness or hopelessness, and participation in schools
- Found that breakfast habits could be linked to not only educational performance but also mental health.

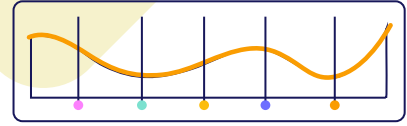


Literature Review cont.

- **The School Nutrition Association**
- Qualitative methods by doing student interviews and focus groups in two public school districts
- Focus was on students' opinions, experiences, and explanations as to why they do or do not eat breakfast.
- Found that students that do eat breakfast had structured routines and easy access to food while
- Students who skipped breakfast had less time, less appetite, or low interest in provided breakfast options, and transportation schedules..

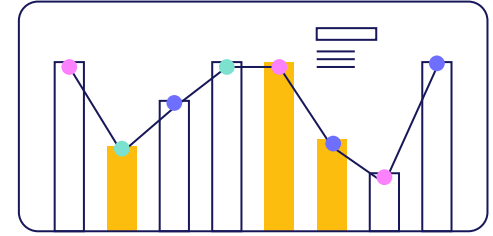


Survey Design



- High school students were randomly selected from multiple classes
- All the highschool classes in the school were written on identical pieces of paper by their room number, folded in half, and placed into a hat.
- We randomly selected 5 pieces of paper, which corresponded to 5 classrooms, where we would randomly select 11 students.
- In the 5 classrooms selected, all the students' names were written on identical pieces of paper, folded, and placed into a hat.
- 11 pieces of paper were drawn from each classroom, resulting in a total sample size of 50 random high school students.
- We showed a QR code to the selected students, which allowed them to fill out an anonymous survey on google forms.

Survey Questions



- **What grade are you in?**

- ☐ 9th
- ☐ 10th
- ☐ 11th
- ☐ 12th

- **Unweighted GPA: _____**

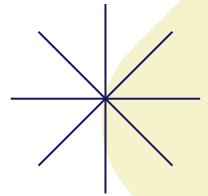
- **Weighted GPA: _____**

- **During a typical school week (Monday–Friday), on how many days do you eat breakfast before school begins?**

- ☐ 0 days
- ☐ 1 day
- ☐ 2 days
- ☐ 3 days
- ☐ 4 days
- ☐ 5 days

"Eats breakfast daily" = 5 days

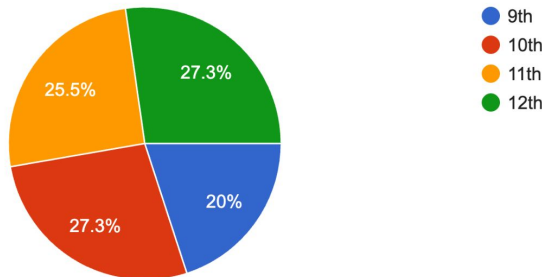
- Success = student selects 5 days
- Failure = anything less than 5 days



Results

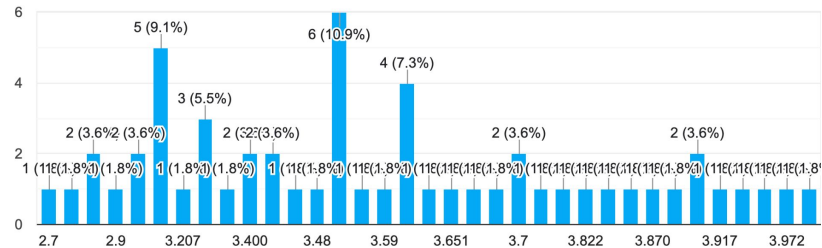
What grade are you in?

55 responses



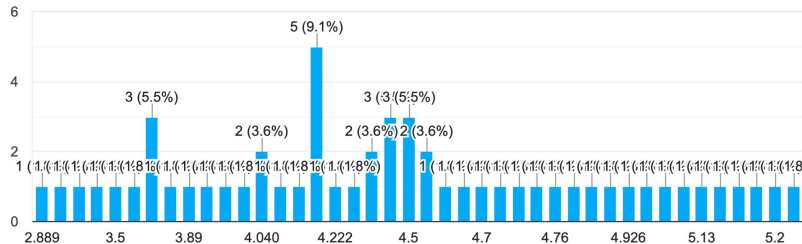
Unweighted GPA (max of 4.0)

55 responses



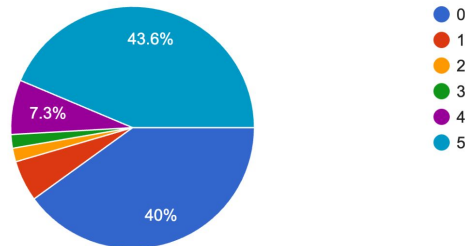
Weighted GPA (max of 6.0)

55 responses



During a typical school week (Monday–Friday), on how many days do you eat breakfast before school begins?

55 responses



Scatterplot

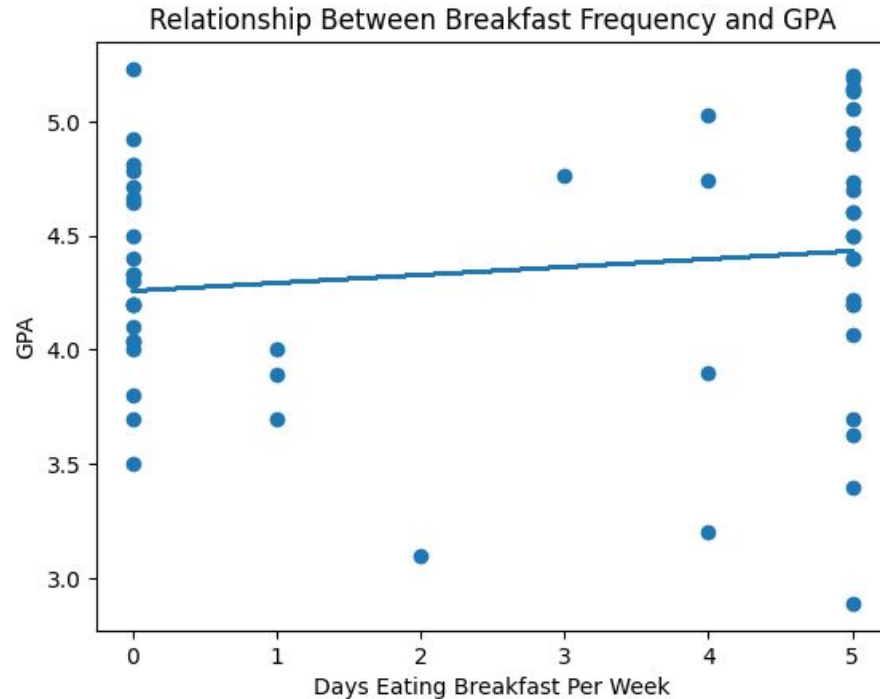
RQ: Is there a relationship between the number of days students eat breakfast per week and their GPA?

R: 0.146

R^2 : 0.021

Regression line:

$$\text{GPA} = 0.0349(\text{Breakfast Days}) + 4.257$$



Relationship Between Variables

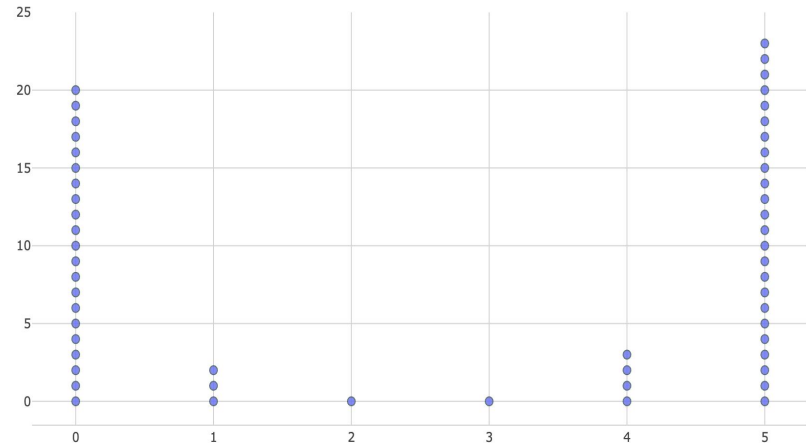
- The scatterplot shows a very weak positive linear relationship between days eating breakfast per week (explanatory variable) and GPA (response variable).
- The linear regression equation is $\text{GPA} = 0.0349(\text{Breakfast Days}) + 4.2575$.
- The slope of 0.0349 means that for each additional day per week a student eats breakfast, the predicted GPA increases by about 0.0349 points on average.
- The data points are widely scattered around the regression line showing a weak correlation.
- Few possible outliers (such as a lower GPA around 2.9 when breakfast is eaten about 5 days per week)
- The trend remains slightly positive meaning that eating breakfast more often is weakly associated with higher GPA, and breakfast frequency alone does not strongly predict GPA.

Descriptive Statistics

- Mean= 2.62 days
- Standard deviation= approx 2.4 days
- Median= 4 days
- Ranged= 0 to 5 days per week.
- Distribution appears to be bimodal
- Cluster at 0 days and another near 5 days

n	55
mean	2.62
median	4
standard dev.	2.368707
minimum	0
maximum	5

Days Students Eat Breakfast in a 5-Day Week



Calculations-Z Test

Breakfast Statistics - Z-Test

communicate:

p = proportion of high school students in Doral Academy High School who eat breakfast daily

$$H_0 = p = 0.274 \text{ (27.4\%)}$$

$$H_a = p > 0.274$$

We will be running a one sample z-test for population proportion

$$n = 55 \quad x = 24$$

$$\hat{p} = \frac{24}{55} = 0.436 \text{ (43.6\%)} \quad \alpha = 0.05$$

conditions:

- We ran a random sample of 55 students, as stated in our survey design
- $55 > 0.10$ (all high school students in Doral Academy)
- $np = 0.274(55) = 15.07 > 10 \checkmark$
- $n(1-p) = 0.726(55) = 39.93 > 10 \checkmark$

Since all conditions are satisfied, we can proceed with the z-test

calculate:

$$P_0 = 0.274$$

$$\hat{p} = 0.436$$

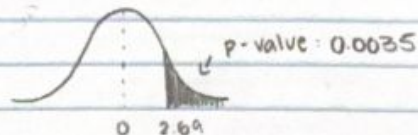
$$n = 55$$

$$\alpha = 0.05$$

$$\text{test statistic} = \frac{\hat{p} - P_0}{\sqrt{\frac{P_0(1-P_0)}{n}}}$$

$$\frac{0.436 - 0.274}{\sqrt{\frac{0.274(0.726)}{55}}} = 2.6937$$

$$P(\hat{p} > 0.274, z > 2.6937) = 0.0035$$



conclude:

Since our p-value (0.0035) is lower than our significance level (0.05), we can reject the null hypothesis. Therefore, there is sufficient evidence that shows the proportion of high school students at Doral Academy that eat breakfast daily is greater than the national average in the US (0.274)

Calculations- Confidence Interval

Breakfast Statistics - Confidence Interval

communicate:

p = proportion of high school students in Doral Academy
High School who eat breakfast daily

We will be running a one sample z interval for
population proportion

$n = 55$ $x = 24$ $Ci = 0.95$

$\hat{p} = 0.436$ $p = 0.274$

conditions:

- we ran a random sample of 55 students, as stated in our survey design
- $55 < 0.10$ (all high school students in Doral Academy)
- $np = 0.274(55) = 15.07 > 10$ ✓
 $n(1-p) = 0.726(55) = 39.93 > 10$ ✓

Since all conditions are satisfied, we can proceed with the confidence interval

calculate:

$\hat{p} = 0.436$

$p = 0.274$

$n = 55$

$z^* = 1.96$

$$\text{confidence interval} = \hat{p} \pm z^* \left(\sqrt{\frac{p(1-p)}{n}} \right)$$

$$0.436 \pm 1.96 \left(\sqrt{\frac{0.274(0.726)}{55}} \right)$$

$$(0.305, 0.567)$$

conclude:

We are 95% confident that the true proportion of high school students in Doral Academy that eat breakfast daily is between 0.305 and 0.567.

Calculations-Chi Square

Chi-Square Test

	o - observed	e - expected		
	Daily (5)	Not Daily (0-4)	total	
9 th	6 4.8	5 6.2	11	Is there an association between grade level and whether students eat breakfast every day?
10 th	4 6.55	11 8.45	15	
11 th	6 6.11	8 7.89	14	
12 th	8 6.55	7 8.45	15	
total	24	31	55	

H_0 : grade level and daily breakfast habits are independent

H_a : grade level and daily breakfast habits are associated

conditions:

- A random sample of students was selected from multiple classrooms at Doral Academy by randomly drawing classroom numbers and student names
- $55 < 0.10$ (all high school students at Doral Academy)
 - ↳ all conditions are satisfied, so we can proceed

calculate:

$$\chi^2 = \sum \frac{(o-e)^2}{e} = \frac{(6-4.8)^2}{4.8} + \frac{(5-6.2)^2}{6.2} + \frac{(4-6.55)^2}{6.55} + \frac{(11-8.45)^2}{8.45}$$

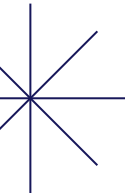
$$df = (4-1)(2-1) = 3$$

$$= 2.8679$$

$$P(\chi^2 > 2.879) = 0.411$$

conclude:

Since our p-value was 0.411 is greater than the significance level of 0.05, we fail to reject the null hypothesis. Therefore, there is not sufficient evidence to conclude that grade level and whether students eat breakfast daily are associated among students at Doral Academy High School



Conclusion



- Proportion z-test: The p-value (0.0035) is lower than our significance level (0.05), so we can reject the null hypothesis. Therefore, there is sufficient evidence that shows the proportion of high school students at Doral Academy that eat breakfast daily is greater than the national average in the US (0.274).
 - Chi-square test for independence: The p-value (0.41) is higher than the significance level (0.05), so we fail to reject the null hypothesis. Based on this, there is not sufficient evidence to conclude that grade level and whether students eat breakfast daily are associated among students at Doral Academy High School.
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